8.5.2.3 *Plunger Pump*

This pump consists of the essential components such as barrel, plunger and two one way spring loaded ball valves etc., The plunger is designed to operate in reciprocating motion inside the barrel. Power required to drive the pump is drawn from the eccentric disc coupled with the cam shaft or by means of a small connecting rod coupled with the crank shaft of the engine. Among the two spring loaded ball valves in the pump, one is connected to the inlet of the barrel whereas the other one is connected to the outlet of the barrel. The figure 8.5.2(c) shows the view of a plunger type oil pump.

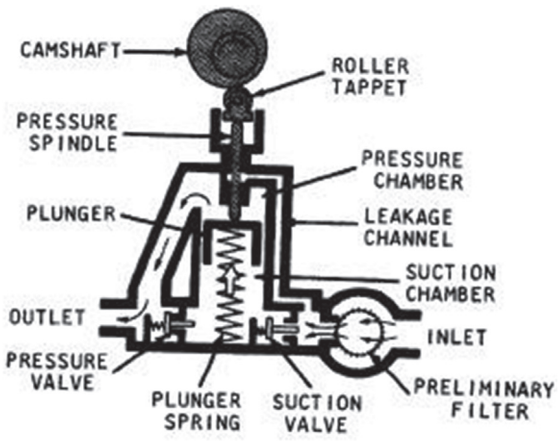


Figure 8.5.2(c) plunger pump

When plunger moves in the upward direction, a partial vacuum is created in the bottom portion of the plunger. So the inlet valve is opened and the oil is sucked and allowed to occupy the vacuum portion. When the plunger moves in downward direction the oil pressure gets increased in the barrel. Now the outlet valve is opened and inlet valve is closed. So the pressurized oil in the pressure chamber is expelled out through the outlet valve. This type of pump is generally not used in automobiles as it does not produce enough amount of pressure. This type of pump is used in stationary oil engines and in few automobile engines.

8.5.2.4 *Vane Pump*

This type of pump has a circular housing. Inside which an eccentric motor is present. The rotor shaft is connected to a skew gear. Surface of the rotor has equally spaced slots, which has many vanes connected. This vanes move to and fro in the slots. Two rings are placed at the center so that the vanes are kept close to the housing. Figure 8.5.2(d) shows the arrangement of the van pump.

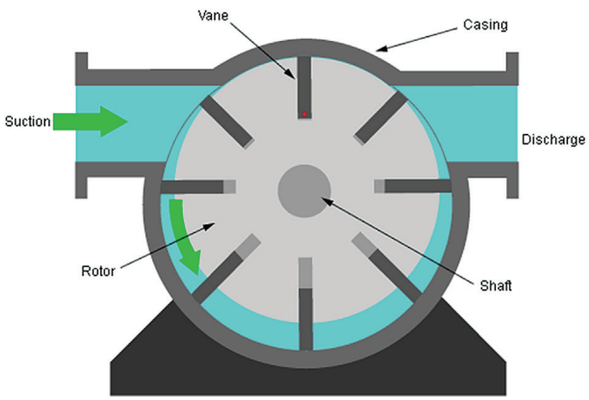


Figure 8.5.2(d) Vane Pump

When the pump is started the shaft rotates and due to the centripetal force vanes are forced outwards. Due to which oil enters into the casing through the inlet. Now the vanes move in the small space between the housing and eccentric rotors. When the vanes move again to the same place the oil in that place moves out through the outlet.

8.6 PRESSURE RELIEF VALVE

The oil pump is designed in such a way to send the fixed amount of oil when

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